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Sublimation of Iodine Crystals Suitable for Profile Angle Measurements

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With 1 Figure

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Very volatile substances that tend to evaporate rapidly at room temperature require special methods to obtain and preserve them in crystalline form long enough to measure angles and to take photographs. Examples of such sublimable substances include *p*-dichlorobenzene, naphthalene, and iodine. The last is chosen to illustrate the method.

Apparatus and Procedure

The apparatus used is described elsewhere[2] and mentioned in the preceding article on pseudosublimation but will be briefly referred to as necessary. The sublimation cell used is a glass ring about 4 mm high and about 16 mm in diameter sandwiched between two microscope slides at their centers. The upper condensing slide is greased with sebum from the face and fingers and the excess is wiped off as completely as possible with a clean, lintless cloth.

At the center of the slide in contact with the surface of the gradient hot plate is placed the ring that forms the cell wall. Within this ring is spread as thinly and uniformly as possible but not as a continuous film, a small quantity of finely crushed iodine. This even distribution is important as it tends to produce an even evolution of vapor and an even build up of symmetrical crystals. Otherwise, distorted forms are more likely to appear. The 50.0° C spot on the gradient hot plate is located by strewing *p*-dichlorobenzene (m. p. 53.0° C) crystals on it. The sublimation cell is placed on the spot where the *p*-dichlorobenzene just fails to melt.

After a few seconds, the ring is capped by the greased condensing slide, preheated to about the same temperature (50.0° C). After heating at this temperature for from one to five minutes with about 1.5 min.

to 2.5 min. as the optimum time, the cell is removed from the hot plate as a unit and allowed to cool to room temperature for a minute or two. The condensing slide is now removed from the cell and immediately covered with a coverslip of much greater area than the crystalline deposit on the slide. A square cover glass of about 25 mm on a side is satisfactory. If the volatile crystals are left in the subliming cell for any length of time at room temperature, they will start to evaporate and lose their sharp outlines. By restricting the space into which they might evaporate by covering

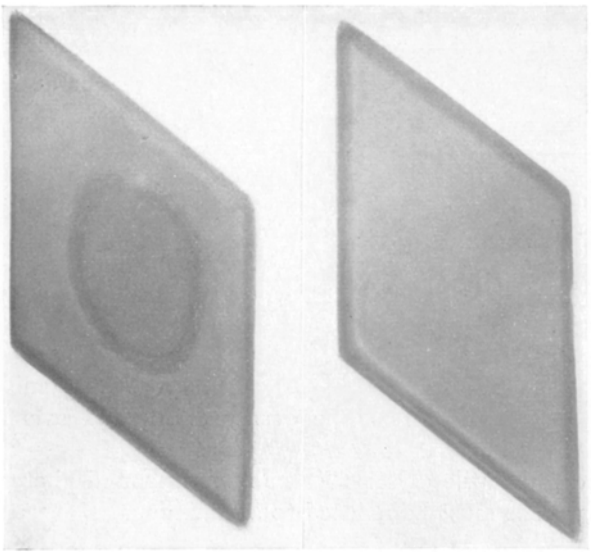

Fig. 1. Iodine Crystals; approx. 450×

the crystals with a cover glass, they are preserved long enough for photographing or angular measurement. Different preparations may have to be made for these purposes. The bottom and top slides can be clamped to the ring of the cell with rubber bands over their ends while the cell cools, and later the cover glass can be sealed to the condensing slide with scotch tape if desired or necessary for more volatile substances than iodine. This is not necessary for iodine unless the same crystals are to be used for both, measurement and photography.

The following values were found for profile angles with straight sides on crystals, the outlines of which were completely in focus[1].

Obtuse Profile Angle of Rhomb (and the two opposite angles of a hexagon): 128.4°, measured average of twelve different angles.

Acute Profile Angle of Rhomb (and the two opposite angles of another hexagon): calculated, 180° — 128.4° = 51.6°; measured: 51.4°, an average of three different angles.

Four Equal Angles of a Hexagon: 115.7°, measured as an average of six different values.

Four Equal Angles of a Hexagon: calculated: 0.25 (720° — 2 × 128.4°) = 115.8°.

Four Equal Angles of Another Hexagon: calculated: 0.25 (720° — 2 × 51.6°) = 154.2°.

Each angle was measured once in each of the four quadrants marked out by the cross hairs in the ocular, and the values were averaged for each result reported. This is expecially important for a black, opaque crystal like iodine since the variation in individual measurements is greater than for a transparent crystal where both sides of a boundry line are visible. The average is satisfactory and can be duplicated even in cases where the difference in individual measurements may amount to as much as two angular degrees or ± 1.0° from the average.

Fig. 1 shows a representative iodine crystal.

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Summary

The sublimation and preservation of very volatile substances are discussed. The material must be evenly distributed over the heated surface of the cell.

To preserve volatile crystals for measurement and photography, the surrounding space must be reduced to a minimum.

Zusammenfassung

Um gleichmäßig entwickelte, plattenförmige Kristalle zur Profilwinkelmessung und Mikrophotographie zu erhalten, ist es notwendig, das Sublimationsgut gleichmäßig auf dem Boden der Kammer zu verteilen. Um die Form der Kristalle flüchtiger Substanzen für einige Zeit unverändert zu erhalten, müssen sie nach der Sublimation schnell in einem kleinen Raum eingeschlossen werden, um die Verdampfung hintanzuhalten.

Résumé

Pour obtenir des cristaux en plaques, régulièrement développés, pour la mesure des angles dièdres et leur microphotographie, il est nécessaire de distribuer bien régulièrement la matière à sublimer sur le fond de la chambre. Pour empêcher pendant un certain temps la forme des cristaux des substances volatiles de se modifier, on doit les enfermer rapidement dans un petit volume après leur sublimation afin de retarder la vaporisation.

Literature

[1] *E. M. Chamot* and *C. W. Mason*, Handbook of Chemical Microscopy, Vol. 1, New York: Wiley. 1938. p. 406.

[2] *A. C. Shead*, Mikrochem. Acta [Wien] **1959**, 657.